

Evaluating Vision-Based Tyre Inspection: Checkpoint Sensitivity, Region Interventions, and Boundary Localisation

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ABSTRACT Vision-based tyre inspection often combines classification, segmentation, and geometric localisation, yet strong component scores need not produce a useful downstream improvement. We present a task-linked evaluation approach connecting checkpoint policy, paired region interventions, and boundary representation. A collection of 418 unique photographs supports grouped internal classification and localisation experiments; a separately allocated 120-image subset supports matched boundary learning. Reanalysis of saved epoch histories shows substantial fold-specific checkpoint sensitivity: MobileNetV4's validation-selected macro-F1 on fold 1 is 0.9924, compared with 0.7393 at epoch 60. Ordinal error metrics and an explicit group-allocation audit complement these scores. Across 81 localisation runs, high region overlap coexists with modest two-pixel boundary agreement. Fixed full/tyre/tread probability fusion changes the frozen classifier's mean macro-F1 by -0.01324, with losses concentrated on one fold rather than uniform across allocations. On the matched 24-image boundary test, point-supervised HRNet and mask-supervised SegFormer obtain horizontal errors of 1.312% and 1.721% of image width, respectively. The 23.75% mean reduction favours point learning on 22 image averages, with complete mask-derived point coverage. Threshold sensitivity and preserved video failures identify further operating limits. The approach connects component scores to downstream utility and makes representation-dependent gains and failures inspectable within a limited-data tyre-inspection setting. Physical wear and alignment accuracy require separate reference measurements.

INDEX TERMS Boundary localisation, explainable artificial intelligence, image segmentation, mileage proxy, reproducibility, tyre inspection.

I. INTRODUCTION

VISION-based tyre inspection must connect a visible feature to a useful decision. Region segmentation, image classification, and boundary localisation are natural components, but a strong component score does not establish that the complete inspection pipeline improves. A segmenter can agree with a large tread region while displacing its boundary; a crop can remove useful context from a classifier; and a validation-selected checkpoint can give a different impression from a fixed training endpoint. These are practical evaluation questions for a system assembled from pretrained models.

This paper studies three questions: *RQ1*, how do checkpoint policy and held-out group affect mileage-proxy recognition? *RQ2*, does predicted localisation improve a frozen full-image classifier through cropping or fixed probability fusion? *RQ3*, how does direct point learning compare with boundaries

extracted from dense masks on matched images? The targets are explicitly separate: ordered mileage categories, manual image regions, and horizontal tread-boundary coordinates. None is substituted for measured tread depth or wheel alignment.

We present a task-linked evaluation approach for vision-based tyre inspection that connects recognition decisions to the spatial evidence used to support them. Its methodological contribution is to evaluate each component change against an explicit reference while preserving the checkpoint, held-out group, decision rule, and prediction coverage. This connects three levels of evidence that are often reported as separate component scores: region agreement, downstream classification change, and boundary-coordinate accuracy. The resulting analysis identifies where an improvement transfers across stages and where it does not.

The contributions are threefold. First, an endpoint-aware recognition analysis combines group-specific macro-F1 and ordinal errors to reveal performance changes concealed by pooled checkpoint summaries. Second, paired crop and probability-fusion interventions quantify the downstream effect of predicted regions on a fixed classifier, alongside area and boundary agreement. Third, a matched point-versus-mask boundary study couples coordinate error with extraction coverage and image-level variation, then preserves rejected geometry during video inspection. Together, these contributions provide a practical basis for selecting and inspecting model components under limited-data conditions.

The study combines 153 retained classification runs, 81 localisation runs, and a three-seed matched boundary experiment. Retrospective analyses reconstruct checkpoint endpoints from saved histories, verify group membership, assess explanation-gate sensitivity, and aggregate boundary differences by image and point location. These diagnostics use the existing experimental cohort. Figure 1 shows how the components and evidence are connected.

II. RELATED WORK AND RESEARCH POSITION

A. TYRE INSPECTION: MEASUREMENT, RECOGNITION, AND DECISIONS

Optical tread inspection predates current deep networks. Nevin and Daoud evaluate laser/camera drive-over measurements against mechanical gauges [1]; Wang et al. use calibrated laser-plane geometry to recover tread depth [2]. Tire-Eye instead uses visible groove geometry and tread-wear indicators as references [3]. RGB-depth point clouds have also been used to measure rubber tread dimensions [4]. These methods have physical scales or reference geometry that a monocular mileage label does not provide.

Recent learning-based studies address several different inspection tasks. Petrović et al. combine Mask R-CNN tread-area segmentation with HOG-based condition classification [5]. Their segmentation-to-classification pipeline makes the downstream value of a predicted region a directly relevant question. Surface-damage inspection also targets local defects rather than mileage categories [6]. Mignot et al. aggregate detected defects using attention-based multiple-instance learning to support a tyre-level repair decision [7]; this illustrates why detected regions and the final decision should be evaluated separately. Vivekanandan and Rajeswari combine visual wear classification with driving-behaviour information and domain adaptation [8]. We do not equate their multimodal condition target with the present mileage-proxy labels or compare headline accuracies across these different datasets.

Wheel alignment is an established measurement problem. Lamy and Basset demonstrate camera-based camber measurement with a physical vehicle prototype [9]. Shi et al. combine rim segmentation, three-dimensional point clouds, and registration to a reference condition for toe/camber estimation [10]. Their use of a geometric reference distinguishes angle measurement from a line or ellipse fitted to an uncalibrated photograph. The alignment software in our artifact is

therefore implementation context, not the article's validated contribution.

Table 1 compares the tasks and evidence requirements of these application studies. The research opportunity motivating our work is the connection between a component's spatial score and its effect on the next decision. Tread-region recognition [5], tyre-level defect aggregation [7], and reference-based alignment [10] establish relevant application pathways. We examine component utility through paired full-frame/crop decisions and matched boundary representations, retaining coverage and failed cases in the interpretation. The distinctive contribution is this joint empirical treatment of transfer between pipeline stages: region overlap, recognition benefit, and coordinate accuracy are evaluated as related but separately testable outcomes.

B. EVALUATION AND REPRESENTATION CHOICES

Shortcut learning motivates testing whether a predictor uses features that transfer beyond its acquisition setting [11]. Leakage analysis and structured-data validation motivate explicit grouping and separation of model selection from evaluation [12], [13]. Our group-allocation audit concerns membership, while checkpoint analysis concerns reuse of validation outcomes; resolving one does not resolve the other.

Residual networks and MobileNetV4 provide contrasting recognition configurations [14], [15]. CORAL supplies an ordinal threshold formulation [16], making class-index error and quadratic weighted kappa useful companions to macro-F1. Grad-CAM and parameter-randomisation checks motivate explanation diagnostics [17], [18], but neither supplies a universal threshold for the project-specific gate. We therefore evaluate threshold sensitivity explicitly.

U-Net, DeepLabV3+, SegFormer, YOLO26, and RT-DETRv2 supply region predictions through differing objectives and implementations [19]–[23]. Boundary-sensitive segmentation evaluation already recognises that area overlap can conceal contour error [24]. Our use of two-pixel boundary F1 is related in motivation but is not the Boundary IoU metric. We test whether this distinction matters for the downstream tyre task.

HRNet preserves high-resolution features [25]. Differentiable coordinate expectation and coordinate classification are also established localisation ideas [26], [27]. Our fixed-row horizontal decoder adapts these principles to the task-specific representation of tread boundaries. The experiment compares a point-supervised HRNet pipeline with a mask-supervised SegFormer pipeline under the same image split and training budget. Confidence calibration and conformal diagnostics [28], [29] are retained in Appendix B, outside the central contribution.

III. DATA AND EXPERIMENTAL DESIGN

A. COLLECTION, LABELS, AND SAMPLING UNIT

The preparation audit starts with 888 raw files and removes 470 byte-identical duplicates, leaving 418 clean RGB photographs at 1152×1536 pixels. Ten pre-generated derivatives

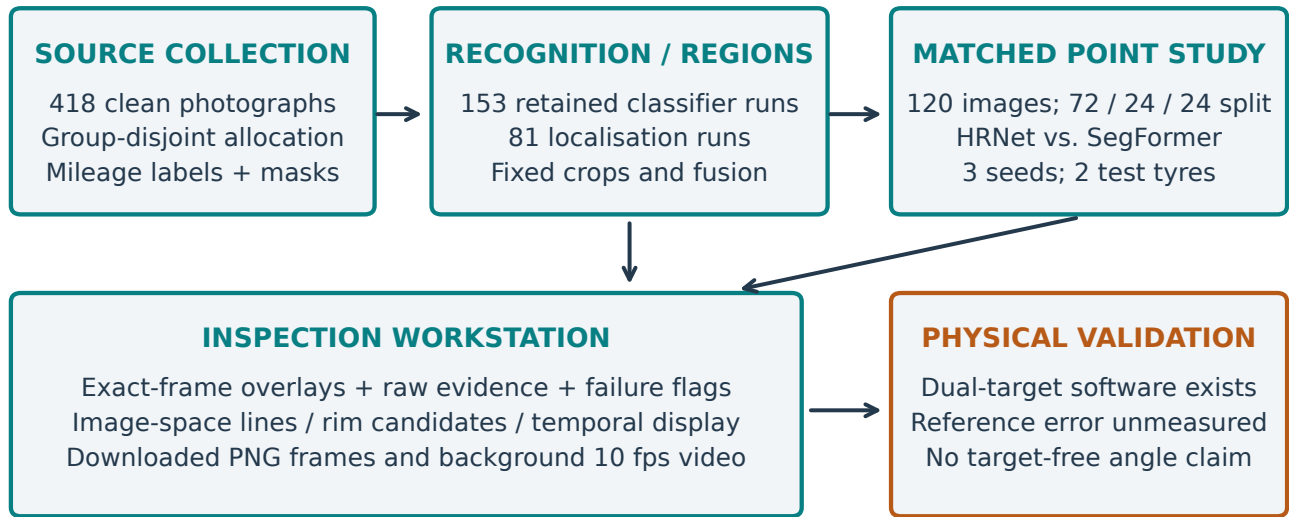


FIGURE 1. Evaluation workflow. Recognition, localisation interventions, and matched boundary learning are the empirical components. The workstation preserves their outputs for inspection. Physical validation is a separate future step; arrows do not imply independent data collections.

TABLE 1. Task-level comparison of closely related inspection studies. NR means that an independent-tyre count or identity-disjoint evaluation could not be established from the consulted primary text; it does not mean absent. Physical validation requirements differ across tasks, so numerical scores are not directly compared.

Study	Input / output	Reference or supervision	Evaluation unit / scope	Relation to this study
Nevin–Daoud (2014) [1]	Laser/camera scans; tread depth	Mechanical depth gauges	Hundreds of tyres in reported comparison	Physical depth measurement
Wang et al. (2019) [2]	Laser-plane images; tread depth	Calibrated geometry; depth gauge	Two tyre types in reported tests	Metric reconstruction rather than proxy labels
TireEye (2022) [3]	Optical groove observation	Tread-wear indicators as scale	Independent cohort count NR	Explicit physical scale
Surface damage (2024) [6]	Local surface images; damage	Image-processing damage features	Independent cohort count NR	Defect task, not mileage recognition
RGB-depth dimensions (2024) [4]	RGB-depth point clouds; dimensions	Physical dimension comparison	Rubber tread measurement setup	Depth sensing supplies geometry
Petrović et al. (2025) [5]	RGB; tread segmentation and condition	Mask R-CNN; HOG features	247 training / 62 test images; tyre count NR	Closest region-to-classification pipeline
Mignot et al. (2025) [7]	Defect observations; repair decision	Tyre-level decision labels	Defects grouped into tyre bags	Global decision after local detection
Vivekanandan–Rajeswari (2026) [8]	Images and behaviour; wear class	Multi-condition classification	Unseen-brand evaluation reported	Multimodal adaptation, distinct target
Lamy–Basset (2010) [9]	Camera observations; camber	Physical prototype / vehicle tests	On-track feasibility evaluation	Reference-based angle measurement
Shi et al. (2026) [10]	Rim masks / 3D points; wheel angles	Zero-angle reference registration	Repeated measurement experiments	Segmentation supports calibrated geometry
Present study	RGB; proxy class / regions / points	Mileage groups, masks, fixed-row points	Grouped internal folds; fixed geometry split	Paired crop interventions, endpoint sensitivity, coverage-aware boundaries

per original produce 4,180 additional files at 768×768 pixels. These derivatives are linked to their source images and are not independent observations. The images were collected manually in a single acquisition session. The limited size and acquisition scope of the dataset restrict the diversity of lighting, viewpoints, and environmental conditions represented.

The low-, mid-, and high-mileage proxy groups contain 169, 97, and 152 clean images, respectively, distributed across 3, 3, and 6 timestamp-derived sessions. Low includes new

tyres and the 100–5,000 mileage group; mid includes the 40,000, 70,000, and 90,000 groups; high corresponds to the 100,000-plus group. These are source-group labels, not measured wear severity. The collection operator subsequently confirmed that the 12 sessions represent 12 distinct physical tyres. This statement is recorded as operator confirmation, without an independent identity audit.

Table 2 uses a three-fold group allocation: every session is assigned to exactly one validation fold. An allocation audit of

TABLE 2. Group-disjoint clean-image allocations. Validation images are reused for internal evaluation; these folds are not an external test set.

Fold	Train	Validation	Interpretation
0	232	186	8 training / 4 validation groups
1	290	128	8 training / 4 validation groups
2	314	104	8 training / 4 validation groups

all 418 image records finds zero train/validation session intersections in each fold. Under the operator’s identity confirmation, these are tyre-group-disjoint internal folds. An earlier appearance-similarity audit flagged the 70,000- and 90,000-mileage sessions as potentially identical, affecting folds 0 and 2; visual resemblance is not proof of identity, and the later operator confirmation states they are different tyres. The recorded groups are therefore disjoint under the supplied identity labels; the earlier visual-similarity flag remains documented in the data provenance.

Classification training can use derivatives of training originals; validation uses clean held-out photographs. The dense-task branch uses clean originals only. Repeated seeds measure optimisation variation on fixed data. Validation-selected results reuse the evaluated fold for checkpoint selection; group disjointness does not make these selected results untouched test estimates.

B. MANUAL REGIONS AND GEOMETRY ANNOTATIONS

The available manually annotated label map M defines overlapping binary regions:

$$Y_{\text{tyre}} = \mathbb{K}[M > 0], \quad Y_{\text{tread}} = \mathbb{K}[M \in \{2, 3\}]. \quad (1)$$

The tread lies within the tyre; treating these labels as mutually exclusive softmax classes would change the task. Bounding boxes are derived from each region. Model agreement with these masks is not a measurement of inter-annotator consistency: no blind independent repeat-annotation study is available.

The geometry extension contains 120 photographs with 720 visible boundary targets. Each image has a left and right tread coordinate on each of three guide rows at 25%, 50%, and 75% of image height. These targets encode horizontal location at a fixed row, not arbitrary landmarks or three-dimensional orientation. The original 12 pilot photographs are excluded from the 120-image package, but the cohort had already informed development. A frozen allocation uses 72 images from eight tyres for training, 24 from two tyres for validation, and 24 from two tyres for testing. Test identities are `mileage_100000_plus` session 006 and `new_tire` session 001. This geometry split is separate from the historical classification folds.

The point annotation interface fixes the three guide rows and records visibility state and horizontal left/right positions. The completion audit verifies 120 complete records and

720 visible coordinates, package/image hashes, and schema consistency. Two near-frame-edge coordinates were specifically inspected. These checks establish data integrity, not inter-annotator reliability: a second blinded annotator and repeated-label error estimate are unavailable. The all-“no visible issue” observation fields are not used as healthy-reference or physical-wear labels. The labels must therefore be interpreted as one annotation protocol rather than an independent metrological standard.

C. ENDPOINTS AND ANALYSIS UNITS

For classification, the validation-selected checkpoint is the earliest epoch attaining the maximum validation quadratic weighted kappa (QWK), using the original strict-improvement rule. The fixed-final checkpoint is epoch 60. We preserve both endpoints. They answer different questions, and selection on the evaluated validation data prevents the larger selected score from being treated as an untouched test result. Localisation and the matched geometry comparison use a fixed epoch-60 endpoint. Unless stated otherwise, numerical differences are descriptive; no population significance claim is inferred from repeated seeds, correlated points, or reused image sets. Missing metrics for a box-only model are undefined rather than zero.

IV. METHODS

A. CLASSIFICATION, IDENTITY AUDIT, AND EXPLANATION SCREENING

The retained classification sweep contains 17 configurations, three folds, and three seeds: 153 runs. Nine additional runs labelled ConvNeXtV2-Small are quarantined because their recorded parameter count and a sampled checkpoint’s tensor signature identify a ResNet-18 substitution. These records remain preserved but are excluded from the architecture comparison. Model names alone are therefore insufficient provenance; implementation, initialisation, head, and checkpoint identity are part of the experimental specification.

Macro-F1 averages the class-specific F1 values over the three proxy classes. For an ordinal model with threshold outputs $q_k(x)$, the implemented prediction is

$$\hat{y}(x) = \sum_{k=0}^1 \mathbb{K}[q_k(x) > 0.5]. \quad (2)$$

We additionally report class-index mean absolute error, $\text{MAE} = N^{-1} \sum_i |\hat{y}_i - y_i|$, with labels 0, 1, 2, and QWK, $1 - \sum_{ab} w_{ab} O_{ab} / \sum_{ab} w_{ab} E_{ab}$, where $w_{ab} = (a - b)^2 / 4$, O is the observed confusion matrix, and E is its chance-agreement counterpart derived from class marginals. These ordinal diagnostics penalise a two-category mistake more than an adjacent error. They are not errors in kilometres or tread depth. Endpoint reconstruction uses all 60 epoch records for two configurations with distinct roles: MobileNetV4 illustrates the checkpoint sensitivity of a high-scoring recognition model, while ResNet-50 supplies the frozen classifier for the region interventions. Each configuration is evaluated over

three folds and three seeds. This diagnostic selection supports interpretation of the existing experiments.

Models trained with a softmax head retain their argmax decision rule. Saved predictions record which rule applies; downstream reconstruction does not replace the ordinal decision with a class-score argmax.

The common classification recipe uses full-frame raw RGB inputs, pretrained full-backbone fine-tuning, a CORAL binary-cross-entropy head, session-balanced sampling, AdamW with initial learning rate 3×10^{-4} and weight decay 0.05, five warm-up epochs, cosine scheduling, and a 60-epoch budget without early stopping. Gradient clipping is 5.0 and training uses half precision. Resolution and batch size follow the architecture registry rather than an identical-resolution assumption: most inputs are 384 pixels, Swin and CoAtNet use 224, and DINOv2 uses 392. Batch sizes range from 16 to 64. Exact per-run configurations and the dataset augmentation policy remain part of the reproducible record.

The explanation screen covers 18 initial candidates and ten shortlist confirmation runs, with 1,208 explanation rows and 35 method-faithfulness rows. A method survives the implemented gate if its randomisation sensitivity exceeds 0.05 and its insertion-minus-deletion area-under-curve value is non-missing. Surviving methods are ranked by that difference. There is no additional requirement that the difference be positive. The threshold is a recorded project heuristic, not a literature-derived validity boundary. We retrospectively repeat the initial-screen gate at 0.03, 0.05, and 0.10 on the same 34 method rows from 17 retained configurations, excluding the quarantined identity. No attribution maps are regenerated and no subsequent models are retrained. Thus the analysis tests initial eligibility sensitivity, not invariance of the complete explanation-guided pipeline.

For a nonnegative saliency map A and tread region T , the normalised tread evidence ratio is

$$\text{TER}_{\text{norm}} = \frac{\sum_{p \in T} A_p / \sum_{p \in \Omega} A_p}{|T|/|\Omega|}, \quad (3)$$

where Ω is the image domain and the ratio is defined only for nonzero denominators. A value above one means concentration beyond the tread's area share. It does not demonstrate causal use of wear features. In this collection, the documented median tread-to-tyre area ratio is 0.990, and 114 images have no visible shoulder. Thus the metric mainly distinguishes tyre from background rather than fine tread from shoulder. Invalid or zero saliency maps remain invalid observations.

The screen selects RegNetY-016, DenseNet-121, and ResNet-50 for one-factor-at-a-time follow-ups. There are 108 discovery runs: three architectures, twelve factor changes, and three seeds on fold 1. A same-fold confirmation stage then evaluates class-weighted sampling, uniform sampling, and random initialisation on ConvNeXtV2-Tiny and MobileNetV4, with three seeds and 60 epochs, giving 18 runs. This tests transfer of factor direction across architectures on the same fold, not independent replication or interactions between separately changed factors.

B. DENSE LOCALISATION AND FROZEN-CLASSIFIER INTERVENTIONS

The localisation branch comprises nine configurations: U-Net/ResNet-34, DeepLabV3+/ResNet-34, SegFormer-B0, SegFormer-B2, YOLO26-n and YOLO26-s in detection and instance-segmentation modes, and RT-DETRv2-R18. Three folds and three seeds produce 81 runs, each trained for 60 epochs on 512-pixel inputs. The declared optimiser is AdamW with learning rate 10^{-4} , weight decay 0.01, and cosine scheduling. Batch sizes are four for semantic and YOLO models and two for RT-DETR.

Semantic models use binary cross-entropy plus soft Dice on two sigmoid channels consistent with (1); the other backends retain their native task losses. Horizontal flipping has probability 0.5 for semantic and repaired YOLO training, while RT-DETR has no online augmentation. YOLO's amended policy disables other hidden augmentations. Final native exponential-moving-average weights are evaluated for YOLO, and final raw weights for the other backends. Differences in pretraining, loss, and backend behaviour are retained as experimental qualifications.

YOLO polygon conversion is verified against the native manual regions. All 836 exported regions meet the required rasterised intersection-over-union (IoU) of at least 0.98; the minimum is approximately 0.980092. This tests the representation conversion, not annotation accuracy. Native masks remain the evaluation reference. For a semantic model, native-resolution sigmoid probabilities are thresholded at 0.5 independently for the tyre and tread channels. Each nonempty binary channel produces one enclosing axis-aligned box and one mask detection; its confidence is the mean probability over threshold-positive pixels. Empty channels produce no detection. These scored class-wise detections enter the same COCO evaluator used for the other backends; they are not a learned object-detection head. Box and mask average precision (AP) use IoU thresholds from 0.50 to 0.95. Region evaluation additionally reports IoU, Dice, and boundary F1 at a two-native-pixel tolerance.

To test downstream usefulness, each localisation run is paired with the frozen epoch-60 ResNet-50 classification checkpoint of the same fold and seed. Five inputs are evaluated: the full frame, predicted tyre crop, predicted tread crop, oracle tyre crop, and oracle tread crop. Crops include 5% padding. A missing predicted region causes a recorded full-frame fallback. The classifier is not retrained on crops; therefore changes can reflect both localisation errors and input-distribution shift.

The fusion analysis uses saved probabilities rather than new training. Five fixed arms are evaluated: full frame, tyre only, tread only, equal tyre/tread fusion, and equal full/tyre/tread fusion. For the last arm,

$$\bar{p} = \frac{1}{3}(p_{\text{full}} + p_{\text{tyre}} + p_{\text{tread}}). \quad (4)$$

Class decisions are reconstructed with the saved head semantics. Each arm is paired with the corresponding full-frame result. There are 405 run/arm metric rows and 56,430

image/arm prediction rows. These counts are not numbers of independent tyres or independently trained fusion systems. Fusion weights are fixed rather than selected for a favourable test result.

C. MATCHED BOUNDARY LEARNING

The HRNet adapter projects its four feature maps to 16 channels each, upsamples them to the highest feature resolution, concatenates them, and produces six coordinate-logit maps. Bilinear sampling at the fixed guide rows yields a horizontal distribution for each point. The target Gaussian has standard deviation 1.5 feature pixels; training minimises mean cross-entropy with the normalised target distribution. Decoding takes the softmax-weighted expected horizontal coordinate, scaled back to native width minus one. Vertical position is prescribed by the guide, not estimated by the network. HRNet-W18 predicts six horizontal coordinates using Gaussian coordinate targets. The matched SegFormer-B0 learns the two overlapping manual mask channels from the same 72 training images; validation/test dense masks do not enter its training loader. Both use input height/width 512/384, batch size two, AdamW with learning rate 10^{-4} and weight decay 0.01, cosine epoch scheduling, no augmentation, frozen batch-normalisation running statistics, three seeds, and epoch 60 for final comparison. Parameter counts are 9,603,962 and 3,714,658, respectively. The supervision and pretrained backbones differ, so a matched split and budget do not isolate architecture as a causal factor.

SegFormer logits are bilinearly interpolated to native resolution and thresholded at sigmoid 0.5. Tread-row extrema supply left/right points; empty rows and frame-clipped extrema are rejected. The declared full-set evaluation replaces a missing point with a training-only mean coordinate and separately reports raw coverage and conditional error. This prevents silent evaluation on an easier detected subset. No replacement is needed in the final comparison.

For image i with native width W_i , target j , and training seed s , the horizontal error is

$$e_{ijs} = \frac{|\hat{x}_{ijs} - x_{ij}|}{W_i - 1}, \quad E_s = \frac{1}{6N} \sum_{i=1}^N \sum_{j=1}^6 e_{ijs}, \quad (5)$$

where $N = 24$ test images. The reported percentage is $100E_s$. Each seed contributes 144 target points, giving 432 paired records across seeds on the same two tyres. Relative error reduction uses the SegFormer mean as denominator, not the number of correctly classified images. The mean over seeds is not an evaluated ensemble.

D. GEOMETRY INTEGRATION

The workstation keeps the analysed frame, portrait orientation, model settings, and predictions together. Deployed geometry uses seed 1, epoch 60, full-image resizing, ImageNet normalisation, and no test-selected rotation search. Local full-precision checks are adapter checks rather than a repetition of the mixed-precision benchmark.

A segmentation mask restricts the search for Canny edge supports. Deterministic consensus and least-squares fits estimate side lines $x = ay + b$ only when support is sufficiently long and consistent. The midpoint line is image-relative. Missing, crossed, clipped, or collapsed boundaries remain visible as rejected hypotheses. A side-view ellipse candidate additionally requires edge support and angular coverage; a detected tyre silhouette is not automatically interpreted as a rim. Temporal smoothing resets at invalid results and discontinuities. It can stabilise a display but cannot correct systematic localisation error.

The separate target-assisted software requires camera intrinsics, a ground-reference target, and a wheel-mounted target, using ChArUco detection [30]. It has synthetic pose tests but no physical reference comparison for the current captures. It is an artifact extension rather than a fourth experimental contribution.

E. EXECUTION AND REPRODUCIBILITY

Kaggle sessions provide training compute and Hugging Face stores persistent records. Jobs retain model/optimiser state, scheduler and scaler state where applicable, random-number-generator state, and execution position. The older dense engine resumes completed epochs; the later geometry engine checkpoints completed optimiser steps and the next-batch cursor. Normal publication is batched at roughly 30-minute intervals, with major completion and catchable interruption requesting a flush. Remote rate limits, failed uploads, or a forced operating-system kill can still lose unpublished progress.

The geometry engine's deterministic resizing repair passed a Tesla T4 resume check with zero maximum parameter difference. This verifies deterministic recovery for the tested interruption path. Immutable revisions and hashes identify the source artifacts, and the manuscript tables and plots are regenerated from these frozen experimental records.

V. RESULTS

A. CLASSIFICATION IS SENSITIVE TO FOLD AND ENDPOINT

Simple baselines reveal substantial fold dependence. The colour probe obtains macro-F1 values of 0.88696, 0.46405, and 0.12322 on folds 0, 1, and 2. The histogram-of-oriented-gradients support-vector machine obtains 0.79602, 0.18182, and 0.98412. These patterns motivate scrutiny of capture-specific cues, but do not identify a unique shortcut mechanism. A historical randomly initialised ResNet-18 trained for 15 epochs averages 0.82135, whereas the later matched nine-run, 60-epoch ResNet-50 control averages 0.82332 at the final endpoint. The two controls differ in backbone and budget and are not interchangeable.

The fold-specific reconstruction in Table 3 reveals information hidden by pooled means. On fold 1, MobileNetV4 changes from validation-selected macro-F1 0.9924 to fixed-final 0.7393, while its fixed-final QWK is 0.9381 and class-index MAE is 0.1094. ResNet-50 changes from 0.8095 to

TABLE 3. Fold-specific checkpoint analysis. Three seeds per cell; selected means the earliest maximum validation QWK. QWK and class-index MAE are reported at epoch 60. MobileNetV4 illustrates recognition endpoint sensitivity; ResNet-50 supplies the frozen downstream classifier.

Model	Fold	Sel. F1	Final F1	Final QWK	MAE
MobileNetV4	0	1.0000	1.0000	1.0000	0.0000
	1	0.9924	0.7393	0.9381	0.1094
	2	1.0000	1.0000	1.0000	0.0000
ResNet-50	0	1.0000	1.0000	1.0000	0.0000
	1	0.8095	0.6159	0.9131	0.1562
	2	1.0000	0.8718	0.9168	0.1058

0.6159 on the same fold, with fixed-final QWK 0.9131 and MAE 0.1563. MobileNetV4 has F1 1.0000 on folds 0 and 2 at both endpoints; ResNet-50's fold-2 F1 falls to 0.8718. These are held-group results within the acquisition cohort, not external validation. The high QWK alongside lower macro-F1 also illustrates why an ordinal task should not be described by one metric.

The full 17-configuration archive is retained alphabetically in Appendix A. Its pooled endpoints are useful for provenance, but the primary analysis preserves folds and explicitly identifies checkpoint selection. The endpoint differences establish sensitivity of the reported conclusion; they do not prove that epoch 60 is the best policy for future data.

B. EXPLANATION DIAGNOSTICS AND SAME-FOLD CONFIRMATION

The original project hypothesis that normalised tread evidence predicts cross-fold stability better than accuracy is not supported: the recorded primary and reference statistics are 0.42710 and 0.90743, respectively, across ten entries. This analysis uses legacy selected-endpoint statistics and is not relabelled as a fixed-budget result. The marking/damage hypothesis is inconclusive or undefined, and the fine-grained architecture hypothesis remains untested because its planned model arms are absent. In particular, fold-1 marking removal operates on zero active marking images; its near-zero effect cannot establish marking independence.

Table 4 reports the confirmation stage. Random initialisation is negative for both architectures at both endpoints. Class-weighted sampling has a positive selected effect for ConvNeXtV2-Tiny but a negative selected effect for MobileNetV4, while both final effects are positive. Uniform sampling is not consistently beneficial. The supported interpretation therefore specifies architecture, endpoint, and the existing fold. The experiment neither isolates factor interactions nor establishes a population-level sampling benefit.

The threshold analysis (Table 5) finds identical initial eligibility and method choices at 0.03 and 0.05: 20 passing method rows across ten models. At 0.10, 16 method rows across eight models remain; RegNetY-016 and MobileNetV4 are excluded. Since RegNetY-016 was one of the subsequent explanation-selected architectures, the downstream selection cannot be described as threshold invariant. We preserve the

TABLE 4. Same-fold confirmation: mean macro-F1 change relative to each matched baseline over three seeds. Selection and final endpoints can give different sampling conclusions.

Architecture	Factor	Δ selected	Δ final
ConvNeXtV2-T	Class-weighted	+0.01257	+0.02396
ConvNeXtV2-T	Uniform	+0.00000	+0.01027
ConvNeXtV2-T	Random init.	-0.42857	-0.38737
MobileNetV4	Class-weighted	-0.02839	+0.01886
MobileNetV4	Uniform	-0.06959	-0.17079
MobileNetV4	Random init.	-0.37150	-0.27525

TABLE 5. Threshold sensitivity of the initial explanation screen: 34 method rows from 17 retained configurations, one seed on fold 1. Thresholds were chosen for this diagnostic; subsequent training was not repeated.

Threshold	Passing methods	Eligible models
0.03	20	10
0.05	20	10
0.10	16	8

original experiment and report this sensitivity instead of retrospectively choosing a gate that gives the preferred shortlist.

C. REGION OVERLAP AND BOUNDARY FIDELITY DIFFER

All 81 localisation runs complete 60 epochs and evaluation, corresponding to 4,860 epoch records. Table 6 restricts the principal summary to fold 1; this retains the original detailed comparison on a challenging held-group fold and does not make fold 1 an external test. Semantic models have tread IoU between approximately 0.972 and 0.976, whereas their two-pixel boundary F1 is approximately 0.358–0.374. YOLO instance-segmentation models obtain tread IoU near 0.856–0.858 and boundary F1 near 0.189–0.226.

Figure 2 makes the metric distinction explicit. Region overlap rewards agreement across a large interior area; boundary F1 is sensitive to contour displacement under its narrow tolerance. Neither metric alone establishes that the model has recovered the information needed for proxy classification, let alone physical tread condition. The evaluation is against one existing manual reference set rather than an independently repeated annotation standard.

D. PREDICTED CROPS AND FIXED FUSION HAVE NEGATIVE MEAN EFFECTS

The paired frozen-classifier analysis does not show an overall gain from the predicted crops. Mean macro-F1 changes are -0.01257 for tyre-only and -0.01467 for tread-only inputs across the 81 localisation source runs. These means do not imply that every configuration/fold pair worsens. They quantify the declared intervention on a classifier trained for full-image inputs, with no crop-specific retraining.

Fixed probability fusion also has negative mean effects (Table 7, Fig. 3). Equal tyre/tread fusion gives -0.01568; equal full/tyre/tread fusion gives -0.01324. The latter is a

TABLE 6. Fold-1 localisation means over three seeds. AP denotes COCO-style AP at IoU thresholds 0.50:0.95. Boundary F1 uses a two-native-pixel tolerance. Dashes denote undefined mask metrics for box-only models.

Model	Box AP	Mask AP	Tread IoU	Boundary F1
DeepLabV3+/R34	0.98612	0.99675	0.97176	0.35988
RT-DETRv2-R18	0.94073	—	—	—
SegFormer-B0	0.91189	0.99836	0.97372	0.37258
SegFormer-B2	0.97657	0.99982	0.97569	0.37412
U-Net/R34	0.91413	0.99704	0.97216	0.35820
YOLO26-n det.	0.95278	—	—	—
YOLO26-n seg.	0.95204	0.95890	0.85815	0.18926
YOLO26-s det.	0.96081	—	—	—
YOLO26-s seg.	0.95227	0.96215	0.85627	0.22623

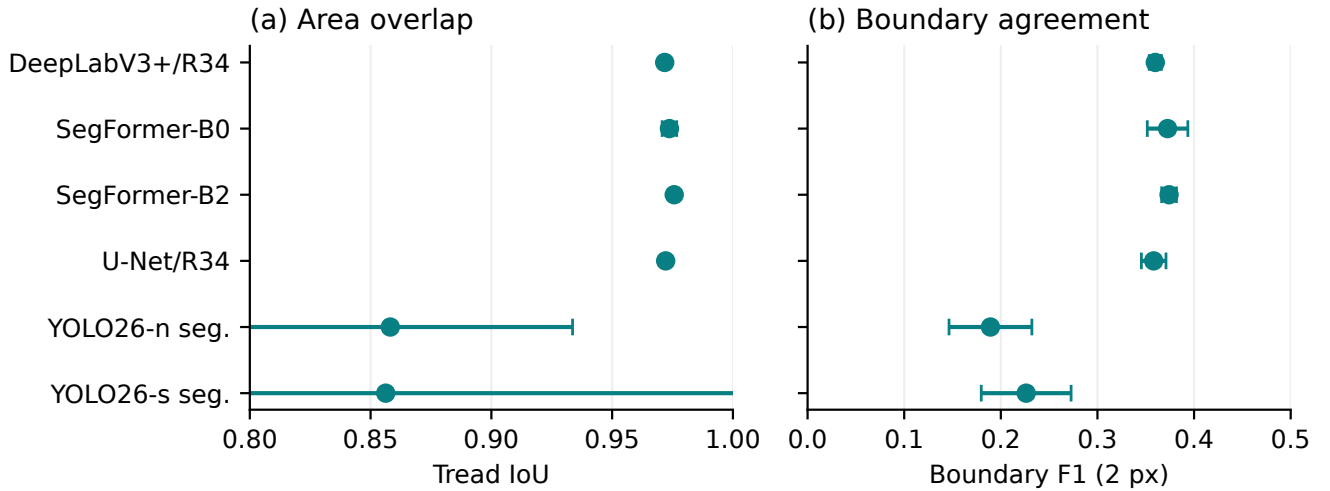


FIGURE 2. Fold-1 semantic/instance-segmentation results: (a) tread IoU and (b) boundary F1 at two native pixels. Points are means and whiskers are sample standard deviations across three training seeds. The panels deliberately use different horizontal scales. Seed variation is not a confidence interval over independent tyres.

TABLE 7. Fixed probability fusion: mean paired macro-F1 changes over 81 source runs. Repeated images and shared classifiers make these rows dependent.

Arm	Rows	Mean Δ
Full frame	81	+0.00000
Tyre only	81	-0.01257
Tread only	81	-0.01467
Tyre + tread	81	-0.01568
Full + tyre + tread	81	-0.01324

decrease of about 1.324 macro-F1 percentage points, not a 1.324% relative error reduction. The result argues against assuming that localisation or additional views automatically help an existing classifier. Lost context, changed scale, crop distribution shift, localisation errors, and correlated predictions are plausible explanations, but the present design does not separate their contributions. Crop-trained recognition and learned fusion would require new experiments. The stratified results in Table 8 show zero change on fold 0 and small positive changes on fold 2. Fold 1 accounts for the negative pooled mean: full/tyre/tread fusion changes F1 by -0.04107

TABLE 8. Paired macro-F1 change by fold; each cell averages nine localisation configurations and three seeds with reused full-image classifiers.

Input	Fold 0	Fold 1	Fold 2
Tyre	+0.00000	-0.04421	+0.00650
Tread	+0.00000	-0.04914	+0.00514
Tyre + tread	+0.00000	-0.05038	+0.00335
Full + tyre + tread	+0.00000	-0.04107	+0.00136

there, compared with +0.00136 on fold 2. Thus the pooled result is not evidence of uniformly harmful cropping.

E. MATCHED POINT LEARNING REDUCES STILL-IMAGE ERROR

Table 9 reports the fixed-final comparison, independently recomputed here from the 432 paired point records. Mean horizontal error is 1.312% of native width for HRNet and 1.721% for matched SegFormer, corresponding to approximately 15.10 and 19.80 native pixels. The absolute difference is 0.409 percentage points of width, or 4.70 pixels. Relative to the SegFormer mean, HRNet reduces error by 23.75%.

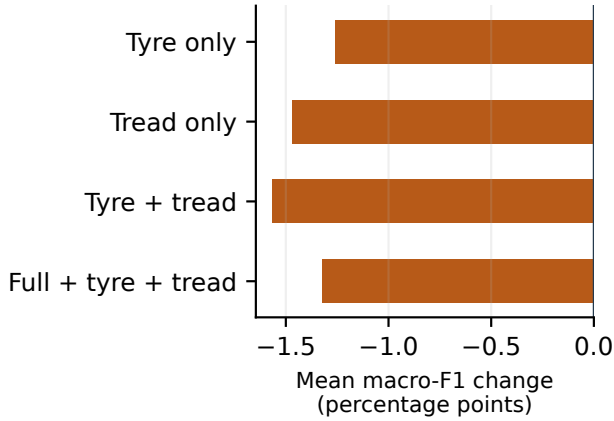


FIGURE 3. Mean paired changes from the full-frame classifier for four fixed crop/fusion arms. Bars average 81 dependent source runs; no independent-sample uncertainty or favourable-arm selection is implied.

TABLE 9. Matched fixed-epoch-60 horizontal point error (% of native width minus one). Each seed uses the same 24 images from two test tyres; coverage is the SegFormer raw point coverage.

Seed	HRNet	SegFormer	Coverage (%)
1	1.391	1.788	100
2	1.284	1.705	100
3	1.261	1.668	100
Mean	1.312	1.721	100

SegFormer raw coverage is 100%, with zero fallback substitutions, so its raw and fallback-assisted means coincide.

HRNet has lower mean error in every seed and on each of the two test tyres (Fig. 4). This does not mean every individual point improves. The new-tyre example has higher error for both models than the high-mileage example, but two tyres cannot establish a class effect. The result supports HRNet as a candidate point-localisation component within this matched study. It does not establish an architecture-only advantage because supervision and initial representations differ, nor does it demonstrate downstream classification improvement or physical-angle accuracy.

A more detailed paired analysis averages six points and three seeds within each image. Point learning has lower mean error on 22 of 24 images; image-level differences (mask-derived minus point error) range from -0.556 to +0.857 percentage points of width (Fig. 5). Table 10 shows that improvement is not spatially uniform: it is 0.029 at the left-middle point and 0.847 at the right-upper point. Every location has a positive mean difference, but two image averages favour the mask-derived pipeline. These retrospective summaries reveal where the aggregate gain occurs; images still share test tyres and are not independent subjects for a population significance test.

TABLE 10. Matched test error by point location (percent of native width minus one). Each row averages the same 24 images and three seeds. Positive difference favours point learning.

Point	Point model	Mask-derived	Δ
left lower	1.096	1.413	+0.317
left middle	1.692	1.721	+0.029
left upper	0.997	1.397	+0.401
right lower	1.756	1.919	+0.162
right middle	1.436	2.132	+0.697
right upper	0.894	1.741	+0.847

F. FAILURE VISIBILITY IN INTEGRATION

The saved still-image and portrait-video examples in Fig. 6 show the intended integration behaviour. A valid still-image result can coexist with crossed point predictions, missing mask-derived boundaries, and disagreement on a video frame. The display preserves raw proposals and withholds invalid widths or connecting geometry. There are no video point labels for a quantitative transfer test, so these examples establish observable failure modes rather than their prevalence.

A local five-call warm inference check on an NVIDIA GTX 1650 measured medians of 72.60 ms for HRNet and 33.48 ms for matched SegFormer, including preprocessing, inference, extraction, and GPU synchronisation. This is a small software check, not a controlled deployment benchmark. The current artifact also exports analysed PNGs and separately processes video at ten output frames per second; this export setting is not an inference-throughput claim. The qualitative examples focus on the predicted boundaries and their rejection states, which are directly relevant to inspecting model output.

VI. DISCUSSION

A. ANSWERS TO THE RESEARCH QUESTIONS

RQ1: checkpoint policy and held-out group. The fold-specific analysis shows that a pooled near-perfect selected score can obscure a materially weaker fixed-final result on one group allocation. Reporting F1 with class-index MAE and QWK makes the type of error more interpretable. Recorded session membership is disjoint under the supplied tyre identities, while checkpoint selection uses the evaluated validation fold. The resulting scores describe internal group-held-out performance with an explicit selection policy.

RQ2: localisation as an intervention. Region agreement does not ensure a downstream gain for a classifier trained on full frames. The fold-specific intervention table localises the overall negative mean: fold 1 worsens, fold 0 is unchanged, and fold 2 has small improvements. The paired design identifies the allocations on which the intervention helps or degrades the existing classifier. A crop-trained classifier or learned fusion could behave differently and requires its own controlled evaluation. Boundary F1 and IoU describe complementary spatial properties, consistent with prior boundary-sensitive evaluation work.

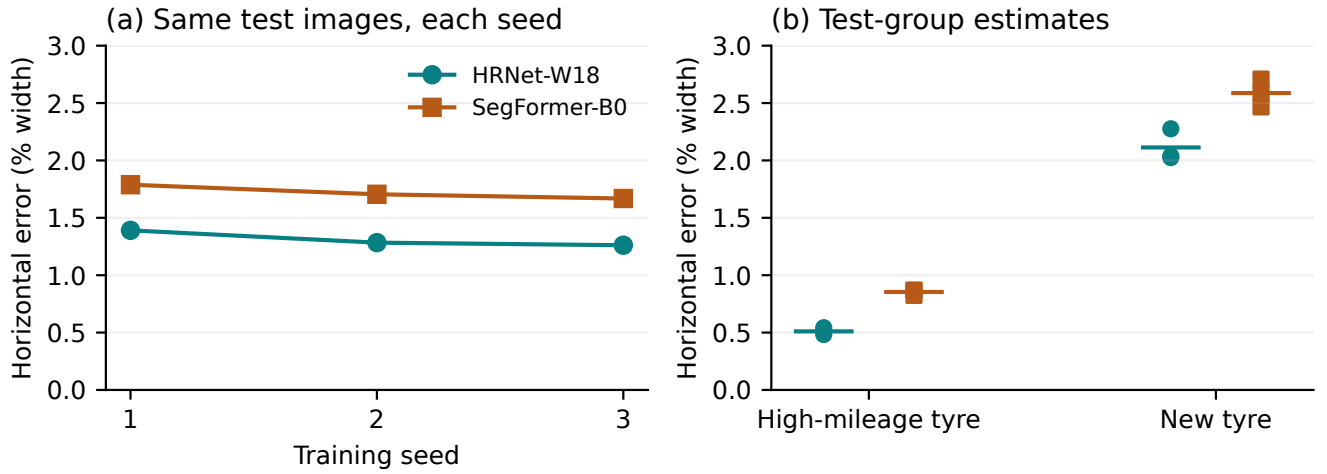


FIGURE 4. Matched boundary comparison. (a) Mean horizontal error for each training seed on the same 24 images. (b) Per-tyre error, retaining all three seed values; horizontal marks are seed means. HRNet is lower in each seed/tyre comparison. These observations are correlated and provide no independent-tyre significance interval.

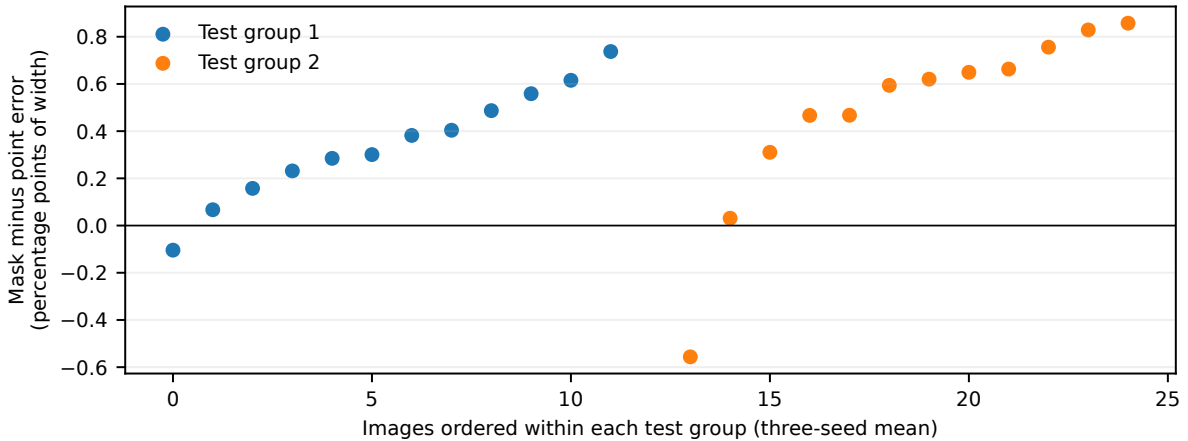


FIGURE 5. Paired image-level boundary differences after averaging the six targets and three seeds. Positive values favour the point-learning pipeline. Images are ordered within each test group for display; their order is not temporal. Two images favour mask-derived boundaries. No confidence interval over new tyres is claimed.

RQ3: boundary representation. Point learning improves the average matched error, while the image-level and location-level analyses show nonuniform benefit. The left-middle location changes little and two image averages favour mask-derived boundaries. Both models remain potentially useful: dense masks provide a search region, while point predictions provide direct coordinate proposals. The study compares complete supervised pipelines; it does not identify the backbone, loss, or pretraining source as the cause of the difference.

B. METHODOLOGICAL CONTRIBUTION AND PRACTICAL IMPLICATIONS

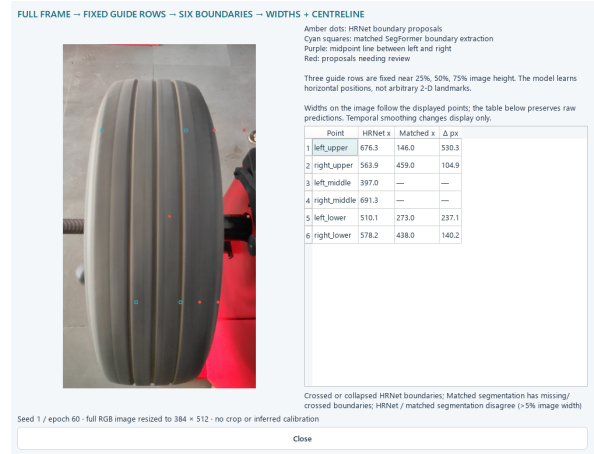
The central contribution is the link between component evaluation and downstream utility. Pairing each region intervention with the same full-image classifier separates its observed effect from changes in classifier training. Reporting both region

overlap and boundary agreement reveals whether spatial quality is concentrated in the interior or at the contour. Comparing point- and mask-derived coordinates on the same allocation then tests the representation used by the geometry stage. This sequence provides complementary evidence for component selection rather than relying on a single accuracy figure.

The findings support concrete engineering decisions. Region predictions should be assessed through their effect on the intended classifier before crop-based inference is adopted. Boundary models should be compared using coordinate error and extraction coverage together. Invalid geometry should remain inspectable even when temporal smoothing is available. The gate-sensitivity result additionally shows that the explanation shortlist depends on its eligibility threshold. These conclusions are supported by the observed interventions and diagnostics; their applicability to other collections can be



(a) Selected internal still-image check



(b) Preserved video-domain failure

FIGURE 6. Qualitative operational evidence, not a new held-out benchmark. Amber HRNet proposals, cyan matched-SegFormer points, and the image-space centreline are shown on the still. The video explanation retains crossed/missing boundaries and disagreement rather than concealing failure. All widths and slopes are image-relative. The images are reproduced from the saved integration checks without altering predicted coordinates.

tested using the same protocol.

C. SCOPE OF THE EVIDENCE

The manually collected dataset supports controlled comparisons within its acquisition scope. Repeated images, seeds, and point coordinates remain correlated. The classification folds cover the available groups but are internal, and the fixed geometry test covers two of those groups. The 432 point records provide repeated measurements within the fixed test cohort. Results are therefore reported descriptively at the seed, image, and tyre levels, without population-level significance claims.

Mileage categories are proxies, not measured tread depth or roadworthiness. Tread design, background, illumination, or capture conditions may contribute to prediction; the baseline variation motivates that concern but does not identify its cause. The available manual labels lack a blinded repeat-annotation estimate. The point-versus-mask comparison differs in supervision, pretrained representations, and capacity. These limits define the applicable conclusions rather than invalidate the observed paired results.

Video examples expose failure but have no labelled prevalence estimate. Local timing checks do not establish sustained throughput. Target-assisted alignment is software with synthetic tests; a real reference-angle experiment is required for physical accuracy. The demonstrated outcomes concern image-space evaluation and inspection behaviour.

D. EXPERIMENTS THAT WOULD ADD INDEPENDENT EVIDENCE

The highest-value next boundary experiment is rotating held-out tyre groups with frozen model settings and fresh training for each split, while retaining a separate development allocation. Each held-out allocation requires models trained exclusively on its corresponding training groups. A shared-backbone point-versus-mask ablation would separate rep-

resentation from backbone more directly. These extensions would test generalisation across allocations and distinguish the effects of representation and model architecture.

A blinded second annotation pass on a stratified subset would estimate label repeatability. A tyre-disjoint labelled video set would permit missing-boundary, crossing, point-error, and temporal-consistency rates. Physical claims require depth or wheel-angle references with repeated mounting and runout assessment. These studies address different uncertainties; adding more architectures on the same split would not substitute for them.

VII. ARTIFACT AVAILABILITY AND REPRODUCIBILITY

Source code and documentation are available at <https://github.com/Shanmuk4622/tyre-wear-alignment-cv>. Persistent study artifacts are at <https://huggingface.co/datasets/Shanmuk4622/tyre-wear-study>. The reporting snapshot is pinned to 22d5a6bc9f953ba3bf2a75919edc7db3193b317b; the HRNet report to a92c0f9c5c1b78c6a06e13d51e18722195230658; and the matched comparison to bbe586c6f00cf12ae4cac8b2e9cb4f875b272abb.

The manuscript package includes numerical source hashes and reproducible analysis scripts. The latter checks split membership, reconstructs endpoints from 18 complete epoch histories, repeats explanation-gate decisions, and aggregates paired boundary and fusion records. The analyses operate on frozen experimental records, allowing each reported summary to be traced to its source.

Relevant dense-task versions include Ultralytics 8.4.20, Transformers 4.51.3, segmentation-models-pytorch 0.5.0, and timm 1.0.15. Local integration uses PyTorch 2.5.1 with CUDA 12.1 and OpenCV 4.11.0. Per-phase environments, data lineage, frozen notebooks, and model identities remain necessary for scientific reproduction; a successful paper build alone is not a replication of training.

VIII. CONCLUSION

This work contributes a task-linked evaluation approach that connects checkpoint policy, localisation interventions, and boundary representation in vision-based tyre inspection. Fold-specific ordinal diagnostics show that pooled selected scores obscure checkpoint and group dependence. Predicted crops and fixed fusion reduce the frozen classifier’s mean F1 overall, with the effect concentrated on one fold. Point learning reduces matched horizontal error by 23.75% relative to mask-derived boundaries and improves 22 of 24 image averages, with complete comparator point coverage. The gain describes the stated pipelines and fixed evaluation cohort. These results support traceable component evaluation and failure visibility; broader transfer and physical measurement remain separate validation tasks.

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DECLARATIONS FOR AUTHOR REVIEW

Proposed contributions (requiring author confirmation): Sreenivasa Reddy Edara: supervision, methodology review, and writing–review and editing. Shanmukesh Bonala: investigation, software, data curation, formal analysis, visualisation, and writing–original draft. These role assignments are a proposed statement, not a verified record of individual work.

Competing interests: author confirmation pending. *Image collection and publication permissions:* author confirmation pending. These fields must be completed before submission.

APPENDIX A ARCHIVAL CLASSIFICATION SUMMARY

Table 11 and Fig. 7 retain the complete original configuration summary. Ordering is alphabetical. These pooled values document endpoint differences; the fold-specific results in the main text remain the primary interpretation.

APPENDIX B CONFIDENCE DIAGNOSTICS

The previously executed confidence analysis averages the saved probabilities from the three explanation-selected architectures and their three seeds within each fold. Each fold’s held-out photographs are stratified into three disjoint image subsets for temperature fitting, conformal calibration, and final diagnostics, using the recorded seed 100 + fold. This subdivision is image-level, not a new tyre-disjoint split. Temperature is fitted by minimising negative log-likelihood on the first subset. Expected calibration error uses 15 equal-width confidence bins. Unlike the standalone ordinal decision in (2),

TABLE 11. Archival classification summary in alphabetical order. Each row averages nine runs (three grouped folds and three seeds). These internal validation summaries are retained for completeness; fold-specific analysis is primary.

Architecture	Selected	Final
CLIP ViT-B/16	0.95076	0.78897
CoAtNet-0	0.80537	0.72041
ConvNeXtV2-T	0.90891	0.90238
DeiT-III-S	0.74726	0.72855
DenseNet-121	0.89660	0.83852
DINOv2-B	0.91277	0.88380
DINOv2-S	0.87346	0.78842
EfficientNetV2-S	0.97615	0.70874
MaxViT-T	0.92189	0.85982
MobileNetV4	0.99746	0.91310
RegNetY-016	0.93495	0.74087
ResNet-50	0.93651	0.82922
ResNeXt-50	0.92260	0.76566
Swin-S	0.95259	0.85744
Swin-T	0.97673	0.83478
VGG16-BN	0.89194	0.45870
ViT-S	0.93235	0.75858

this saved ensemble diagnostic explicitly uses the argmax of its averaged class probabilities.

Conformal nonconformity is $1 - p_y$ on the separate calibration subset. The implementation uses the higher empirical quantile at $\min\{1, \lceil (n_{cal} + 1)0.9 \rceil / n_{cal}\}$ and includes test class k when $1 - p_k$ does not exceed that threshold. Coverage and mean set size are then measured on the third subset. The recorded abstention rate counts sets larger than one, leaving empty sets outside that field. We preserve this implementation detail rather than interpreting the field as a complete reject-option metric.

For CORAL outputs $q_0 = \sigma(z_0)$ and $q_1 = \sigma(z_1)$, the saved implementation constructs $r = (1 - q_0, q_0 - q_1, q_1)$ and normalises $p_k = \max(r_k, 10^{-8}) / \sum_j \max(r_j, 10^{-8})$. Clamping guarantees a valid probability vector even if cumulative outputs are out of order; it does not guarantee calibrated probabilities. Averaging these vectors and taking argmax can differ from counting thresholds in (2). The uncertainty analysis is consequently a separate decision pipeline, not interchangeable with the reported ordinal classifier.

A. RECORDED OUTCOMES

Temperature scaling leaves the recorded macro-F1 unchanged and reduces confidence-error metrics on the retained calibration-test subsets (Table 12). On fold 1, expected calibration error decreases from 0.18955 to 0.07377, and negative log-likelihood from 0.35630 to 0.16226, with 41 test images. Extremely small calibrated errors on the other folds are not evidence of deployment-grade certainty.

At nominal 90% coverage, the observed conformal coverages are 0.86885, 0.97561, and 0.93939 on folds 0, 1, and 2, respectively (Table 13). Thus only fold 0 is below nominal in these observed subsets. Mean set sizes below one on folds 0 and 2 imply empty sets. The recorded abstention field does

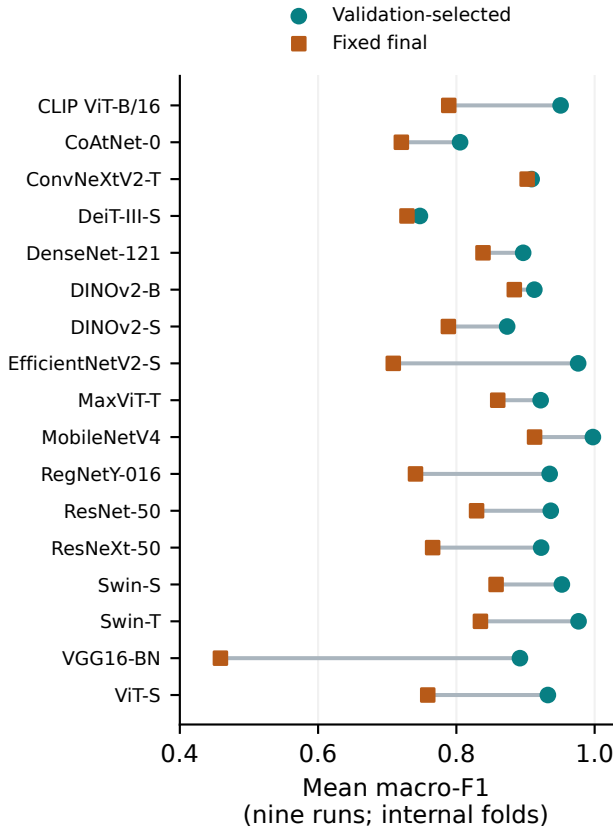


FIGURE 7. Validation-selected and fixed-final macro-F1 means for all 17 retained configurations. Each point averages nine runs on the same historical folds; connecting segments highlight endpoint differences and are not uncertainty intervals. Configurations are alphabetical, not performance-ranked.

TABLE 12. Confidence diagnostics on held-out calibration-test subsets. ECE is the recorded expected calibration error; values near zero on internal folds do not imply external reliability.

Fold	Scores	n	F1	ECE	NLL
0	Raw	61	1.0000	0.2115	0.2425
0	Temp.	61	1.0000	0.0000	0.0000
1	Raw	41	0.8030	0.1896	0.3563
1	Temp.	41	0.8030	0.0738	0.1623
2	Raw	33	1.0000	0.1293	0.1438
2	Temp.	33	1.0000	0.0000	0.0000

not count all empty-set events and cannot be presented as a complete rejection policy. Small subset sizes, dependent acquisition, and reuse of internal evidence constrain generalisation of these diagnostics.

TABLE 13. Prediction-set diagnostics at nominal 90% coverage. The recorded abstention field does not count every empty-set event.

Fold	n_{cal}	n_{test}	Coverage	Set size	Abstain
0	62	61	0.8689	0.8689	0.0000
1	43	41	0.9756	1.0976	0.0976
2	35	33	0.9394	0.9394	0.0000

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